ISC Class XII Physics

Chapter 4
Moving Charges and Magnetism

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51 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

I. Biot–Savart's Law and Ampere's Circuital Law
II. Magnetic Force on a Moving Charge
III. Path of a Charged Particle in a Uniform Magnetic Field
IV. Magnetic Force on a Current-Carrying Conductor
V. Force between Parallel Straight Current-Carrying Conductors
VI. Cyclotron
VII. Torque on a Current-Carrying Coil in a Magnetic Field
VIII. Moving Coil Galvanometer and Voltage/Current Sensitivities
IX. Conversion of Galvanometer into Ammeter/Voltmeter

Compiled Answer Key – Step-by-step ISC board-style solutions

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Formula Sheet

Constants Used Throughout This Chapter

$$\mu_0 = 4\pi \times 10^{-7}\ \text{T m A}^{-1} \qquad e = 1.6 \times 10^{-19}\ \text{C}$$

$$m_e = 9.1 \times 10^{-31}\ \text{kg} \qquad m_p = 1.67 \times 10^{-27}\ \text{kg}$$

$$m_\alpha = 6.7 \times 10^{-27}\ \text{kg} \qquad g = 9.8\ \text{m s}^{-2}$$

μ_0 = permeability of free space; e = elementary charge; m_e, m_p, m_α = rest masses of electron, proton, α-particle.

I. Magnetic Field due to Current (Biot–Savart & Ampere)

Field due to infinite straight wire

$$B = \frac{\mu_0 I}{2\pi r}$$

I = current, r = perpendicular distance from wire.

Field at centre of a circular coil

$$B = \frac{\mu_0 N I}{2r}$$

N = number of turns, r = radius of coil.

Field on the axis of a circular loop

$$B = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$$

a = radius of loop, x = distance of point from centre along axis.

Field inside a long solenoid

$$B = \mu_0 n I$$

n = number of turns per unit length = N/L.

Field due to a moving point charge

$$B = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$$

q = charge, v = speed, θ = angle between v and line joining charge to point.

II. Magnetic Force on a Moving Charge

Lorentz magnetic force

$$F = qvB\sin\theta$$

q = charge, v = speed, B = field, θ = angle between $\vec{v}$ and $\vec{B}$.

III. Path of a Charged Particle in a Uniform Field

Radius of circular path

$$r = \frac{mv}{qB}$$

m = mass, v = speed (perpendicular to B).

Time period & frequency of revolution

$$T = \frac{2\pi m}{qB}, \qquad f = \frac{qB}{2\pi m}$$

Independent of the speed v of the particle.

Speed from accelerating potential

$$v = \sqrt{\frac{2qV}{m}}$$

V = accelerating potential difference.

Oblique entry (helical/spiral path)

$$r' = \frac{mv}{qB\sin\theta}$$

θ = angle between velocity and field; $\theta = 90^\circ$ gives circular path.

IV. Magnetic Force on a Current-Carrying Conductor

Force on a current-carrying conductor

$$F = BIL\sin\theta$$

L = length of conductor, θ = angle between current direction and B.

Microscopic (drift velocity) form

$$I = nAv_d e$$

n = free electron density, A = area of cross-section, v_d = drift velocity.

V. Force between Parallel Straight Current-Carrying Conductors

Force per unit length between two wires

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

d = separation between wires; like currents attract, unlike currents repel.

VI. Cyclotron

Cyclotron magnetic field

$$B = \frac{2\pi m f}{q}$$

f = frequency of the applied oscillating voltage (= revolution frequency).

Maximum velocity & energy

$$v_{max} = 2\pi f R, \qquad E_{max} = \tfrac{1}{2} m v_{max}^2$$

R = radius of the dees.

VII. Torque on a Current-Carrying Coil in a Magnetic Field

Torque on a current loop

$$\tau = NBIA \sin\theta$$

A = area of coil, θ = angle between the normal to coil and $\vec{B}$.

Magnetic dipole moment

$$m = NIA$$

Torque may also be written $\tau = mB\sin\theta$; maximum torque $\tau_{max} = mB$ when $\theta = 90°$.

VIII. Moving Coil Galvanometer: Sensitivities

Current sensitivity

$$I_s = \frac{NBA}{k}$$

k = torsional constant of the suspension.

Voltage sensitivity

$$V_s = \frac{I_s}{R} = \frac{NBA}{kR}$$

R = resistance of the galvanometer coil.

IX. Conversion of Galvanometer into Ammeter/Voltmeter

Shunt resistance (galvanometer → ammeter)

$$S = \frac{I_g R_g}{I - I_g}$$

R_g = galvanometer resistance, I_g = full-scale deflection current, I = desired ammeter range.

Series resistance (galvanometer → voltmeter)

$$R = \frac{V}{I_g} - R_g$$

V = desired voltmeter range.

Net (effective) resistance

$$R_{ammeter} = \frac{R_g S}{R_g + S}, \qquad R_{voltmeter} = R_g + R$$

Ammeter resistance is always very small; voltmeter resistance is always very large.

I. Magnetic Field due to Current: Biot–Savart's Law and Ampere's Circuital Law

Question 1

A long straight wire in a horizontal plane carries a current of 50 A in the north to south direction. Find the magnitude and direction of the magnetic field at a point 2.5 m east of the wire.

★ **Likely Exam Question**

Given: $I = 50$ A, $r = 2.5$ m

Formula: $B = \dfrac{\mu_0 I}{2\pi r}$

Solution:

$$B = \frac{4\pi \times 10^{-7} \times 50}{2\pi \times 2.5} = \frac{2 \times 10^{-7} \times 50}{2.5} = 4.0 \times 10^{-6} \text{ T}$$

Direction: By the right-hand thumb rule, curl the fingers in the direction the field circles the wire while the thumb points along the current (north to south). At a point east of the wire, the field points *vertically upward*.

Final Answer

$B = 4.0 \times 10^{-6}$ T, directed vertically upwards.

Common Mistake: Students often forget that the direction of B must be found using the right-hand thumb rule about the wire, not about the observation point.

Question 2

An electron-beam carries a current of 5 microampere. Calculate: (i) the number of electrons passing through a point per second, (ii) the magnetic field produced at a distance of 50 cm.

Given: $I = 5\ \mu\text{A} = 5 \times 10^{-6}$ A, $r = 50$ cm $= 0.5$ m

Formula: $I = ne \Rightarrow n = \dfrac{I}{e}$; $\quad B = \dfrac{\mu_0 I}{2\pi r}$

Solution:

(i) $n = \dfrac{5 \times 10^{-6}}{1.6 \times 10^{-19}} = 3.125 \times 10^{13}$ electrons/second

(ii) $B = \dfrac{2 \times 10^{-7} \times 5 \times 10^{-6}}{0.5} = 2.0 \times 10^{-12}$ T

Final Answer

(i) $n = 3.125 \times 10^{13}$ electrons/s (ii) $B = 2.0 \times 10^{-12}$ T

Question 3

Two long, parallel wires are placed at a distance of 16 cm from each other in air. Each wire has a current of 4 A. Calculate the field B at the mid-point between them when the currents in them are (i) in the same direction, (ii) in opposite directions.

★ **Likely Exam Question**

Given: $d = 16$ cm, so $r = 8$ cm $= 0.08$ m from each wire; $I_1 = I_2 = 4$ A

Formula: $B = \dfrac{\mu_0 I}{2\pi r}$, then combine the two fields at the mid-point vectorially.

Solution:

Field due to each wire at the mid-point:

$$B_1 = B_2 = \frac{2 \times 10^{-7} \times 4}{0.08} = 1.0 \times 10^{-5} \text{ T}$$

(i) When the currents are in the *same* direction, the two fields at the mid-point point in *opposite* senses (one into the page, one out of the page) and exactly cancel.

(ii) When the currents are in *opposite* directions, the two fields at the mid-point point in the *same* sense and add up.

$$B = B_1 + B_2 = 2 \times 10^{-5} \text{ T}$$

Final Answer

(i) $B = 0$ (ii) $B = 2 \times 10^{-5}$ T

Common Mistake: A very common error is to assume same-direction currents always add their fields – at the midpoint between two wires it is the opposite: same-direction currents cancel, opposite-direction currents add.

Question 4

A circular coil of wire having 100 turns, each of radius 8.0 cm, carries a current of 0.40 A. Find the magnetic field at the centre of the coil.

Given: $N = 100$, $r = 8.0$ cm $= 0.08$ m, $I = 0.40$ A

Formula: $B = \dfrac{\mu_0 N I}{2r}$

Solution:

$$B = \frac{4\pi \times 10^{-7} \times 100 \times 0.40}{2 \times 0.08} = \frac{4\pi \times 10^{-7} \times 40}{0.16} = 1000\pi \times 10^{-7}$$
$$B = 3.14 \times 10^{-4} \text{ T}$$

Final Answer

$B = 3.14 \times 10^{-4}$ T

Question 5

An alpha-particle (charge $2e$) moves along a circular path of radius 1.0 Å with a uniform speed of 2.0×10^6 m s^{-1}. Calculate the magnetic field produced at the centre of the circular path.

Given: $q = 2e = 3.2 \times 10^{-19}$ C, $r = 1.0$ Å $= 1.0 \times 10^{-10}$ m, $v = 2.0 \times 10^6$ m s^{-1}

Formula: The moving charge is equivalent to a tiny current loop, $I = \dfrac{q}{T} = \dfrac{qv}{2\pi r}$, and $B = \dfrac{\mu_0 I}{2r} = \dfrac{\mu_0 q v}{4\pi r^2}$

Solution:

$$B = \frac{4\pi \times 10^{-7} \times 3.2 \times 10^{-19} \times 2.0 \times 10^6}{4\pi \times (1.0 \times 10^{-10})^2} = \frac{3.2 \times 10^{-19} \times 2.0 \times 10^6 \times 10^{-7}}{1.0 \times 10^{-20}}$$
$$B = 6.4 \text{ T}$$

Final Answer

$B = 6.4$ T

Common Mistake: Do not forget the alpha-particle's charge is $2e$, not e – omitting this factor of 2 is a very common slip.

Question 6

An air-solenoid has 500 turns of wire in its 40 cm length. If the current in the wire is 1.0 A, find the magnetic field at the axis inside the solenoid.

Given: $N = 500$, $L = 40$ cm $= 0.4$ m, $I = 1.0$ A

Formula: $n = \dfrac{N}{L}$; $B = \mu_0 n I$

Solution:

$$n = \frac{500}{0.4} = 1250 \text{ turns/m}$$
$$B = 4\pi \times 10^{-7} \times 1250 \times 1.0 = 1.57 \times 10^{-3} \text{ T}$$

Final Answer

$B = 1.57 \times 10^{-3}$ T

Question 7

A 80 cm long solenoid of diameter 1.8 cm has 5 layers of windings of 400 turns each and carries a current of 8.0 A. Estimate the magnetic field inside the solenoid near the centre.

Given: $L = 80 \text{ cm} = 0.8 \text{ m}$, total turns $N = 5 \times 400 = 2000$, $I = 8.0$ A

Formula: $n = \frac{N}{L}$; $B = \mu_0 n I$

Solution:

$$n = \frac{2000}{0.8} = 2500 \text{ turns/m}$$
$$B = 4\pi \times 10^{-7} \times 2500 \times 8.0 = 2.5 \times 10^{-2} \text{ T}$$

Final Answer

$B = 2.5 \times 10^{-2}$ T

Common Mistake: The diameter of the solenoid is irrelevant to the field magnitude here (only n and I matter) – it is given only as a distractor.

Question 8

The magnetic field at the centre of a 50 cm long solenoid is 4.0×10^{-2} T when a current of 8.0 A flows through it. What is the number of turns in the solenoid? (Take $\pi = 3.14$.)

Given: $L = 0.5$ m, $B = 4.0 \times 10^{-2}$ T, $I = 8.0$ A

Formula: $B = \mu_0 n I \Rightarrow n = \frac{B}{\mu_0 I}$; $N = nL$

Solution:

$$n = \frac{4.0 \times 10^{-2}}{4 \times 3.14 \times 10^{-7} \times 8.0} = \frac{4.0 \times 10^{-2}}{1.005 \times 10^{-5}} \approx 3979 \text{ turns/m}$$
$$N = 3979 \times 0.5 \approx 1990$$

Final Answer

$N \approx 1990$ turns

Question 9

In the Bohr model of the hydrogen atom, the electron circulates around the nucleus in a path of radius 5.1×10^{-11} m at a frequency 6.8×10^{15} s^{-1}. Find the magnetic field set up at the centre of the orbit.

★ **Likely Exam Question**

Given: $r = 5.1 \times 10^{-11}$ m, $f = 6.8 \times 10^{15}$ Hz

Formula: Equivalent current $I = ef$; $B = \dfrac{\mu_0 I}{2r}$

Solution:

$$I = 1.6 \times 10^{-19} \times 6.8 \times 10^{15} = 1.088 \times 10^{-3} \text{ A}$$

$$B = \frac{4\pi \times 10^{-7} \times 1.088 \times 10^{-3}}{2 \times 5.1 \times 10^{-11}} = 13.4 \text{ T}$$

Final Answer

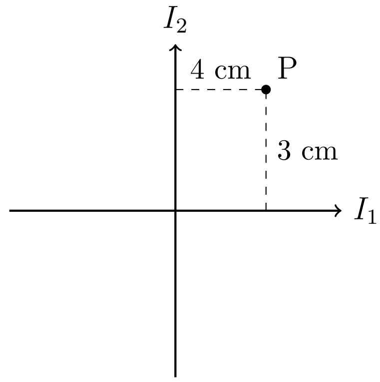
$B = 13.4$ T

Common Mistake: Students sometimes confuse the orbital *frequency* f (revolutions per second) with angular frequency ω. Here $I = ef$ directly, since one electron passes a fixed point f times per second.

Question 10

Two infinitely long insulated wires are kept perpendicular to each other. They carry currents $I_1 = 2$ A and $I_2 = 1.5$ A. (i) Find the magnitude and direction of the magnetic field at point P, which lies 4 cm from the wire carrying I_1 and 3 cm from the wire carrying I_2 (both wires and P lying in the plane of the paper). (ii) If the direction of current be reversed in one of the wires, what would be the magnitude of the field?

★ **Likely Exam Question**



Two perpendicular wires carrying I_1 and I_2, with P at 4 cm from I_1 and 3 cm from I_2

Given: $I_1 = 2$ A at $r_1 = 4$ cm $= 0.04$ m; $I_2 = 1.5$ A at $r_2 = 3$ cm $= 0.03$ m

Formula: $B = \dfrac{\mu_0 I}{2\pi r}$ for each wire; since both wires and P lie in the plane of the paper, both fields at P are perpendicular to the paper and so add/subtract as scalars.

Solution:

$$B_1 = \frac{2 \times 10^{-7} \times 2}{0.04} = 1.0 \times 10^{-5} \text{ T}$$
$$B_2 = \frac{2 \times 10^{-7} \times 1.5}{0.03} = 1.0 \times 10^{-5} \text{ T}$$

(i) With the given current directions both fields point into the plane of paper at P, so they add:

$$B = B_1 + B_2 = 2 \times 10^{-5} \text{ T}$$

(ii) Reversing the current in one wire reverses its field direction at P, so the two fields now cancel exactly (they are equal in magnitude):

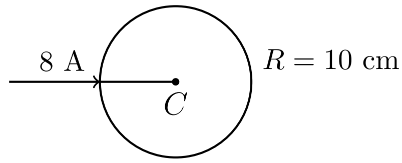
$$B = B_1 - B_2 = 0$$

Final Answer

(i) $B = 2 \times 10^{-5}$ T, perpendicular to the plane of paper, pointing downwards (ii) $B = 0$

Question 11

A long, straight wire is turned into a loop of radius $R = 10$ cm, as shown, with the rest of the wire remaining straight. If a current of 8 A is passed through the loop, find the magnetic field at the centre C of the loop.



Straight wire turned into a loop of radius $R = 10$ cm carrying current 8 A

Given: $R = 10 \text{ cm} = 0.1 \text{ m}, \quad I = 8 \text{ A}$

Formula: Field due to straight (unlooped) part: $B_1 = \dfrac{\mu_0 I}{2\pi R}$; Field due to circular loop: $B_2 = \dfrac{\mu_0 I}{2R}$ – these point in opposite senses at C.

Solution:

$$B_1 = \frac{2 \times 10^{-7} \times 8}{0.1} = 1.6 \times 10^{-5} \text{ T (perpendicular to paper, downwards)}$$

$$B_2 = \frac{4\pi \times 10^{-7} \times 8}{2 \times 0.1} = 5.03 \times 10^{-5} \text{ T (perpendicular to paper, upwards)}$$

Since B_2 and B_1 oppose:

$$B = B_2 - B_1 = 5.03 \times 10^{-5} - 1.6 \times 10^{-5} = 3.4 \times 10^{-5} \text{ T}$$

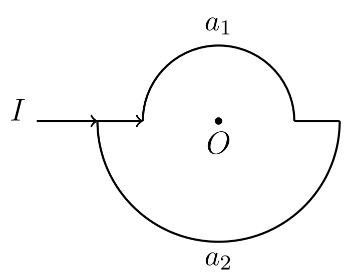
Final Answer

$B = 3.4 \times 10^{-5}$ T, perpendicular to the plane of paper, pointing upwards.

Common Mistake: A frequent error is to simply add B_1 and B_2; because the straight-wire field and loop field act in *opposite* senses at C, they must be subtracted.

Question 12

The figure shows two semi-circular current-loops of radii a_1 and a_2, joined at a common centre O and carrying the same current I. Find the magnitude and direction of the magnetic field at the common centre O.



Two semicircular current loops of radii a_1 and a_2 sharing a common centre O

Given: Semicircular arcs of radii a_1 and a_2 carrying current I, sharing centre O.

Formula: Field at centre due to a full circular loop of radius a: $B = \dfrac{\mu_0 I}{2a}$. A semicircular arc produces exactly half of this.

Solution:

Field due to semicircle of radius a_1: $\dfrac{1}{2}\left(\dfrac{\mu_0 I}{2a_1}\right) = \dfrac{\mu_0 I}{4a_1}$

Field due to semicircle of radius a_2: $\dfrac{\mu_0 I}{4a_2}$

Both fields point in the same sense at O (same current sense around both arcs), so they add:

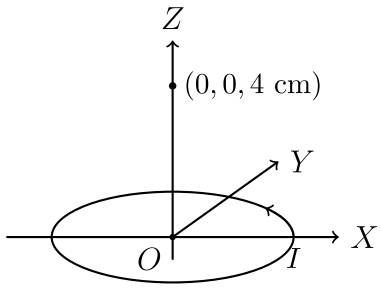
$$B = \frac{\mu_0 I}{4}\left(\frac{1}{a_1} + \frac{1}{a_2}\right)$$

Final Answer

$B = \dfrac{\mu_0 I}{4}\left(\dfrac{1}{a_1} + \dfrac{1}{a_2}\right)$, perpendicular to the plane of paper, directed downwards.

Question 13

Current $I = 2.5$ A flows along the circle $x^2 + y^2 = 9$ cm^2 (with x, y in cm), as shown. Find the magnitude and direction of the magnetic field (in Tesla) at the point $(0, 0, 4$ cm$)$.



Current loop $x^2 + y^2 = 9$ cm^2 in the xy-plane; field required at $(0, 0, 4$ cm$)$ on the Z-axis

Given: Radius of loop $a = 3$ cm, current $I = 2.5$ A, axial distance $x = 4$ cm

Formula: $B = \dfrac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$

Solution:

$$B = \frac{4\pi \times 10^{-7} \times 2.5 \times (3 \times 10^{-2})^2}{2 \times [(3 \times 10^{-2})^2 + (4 \times 10^{-2})^2]^{3/2}} = \frac{4\pi \times 10^{-7} \times 2.5 \times 9 \times 10^{-4}}{2 \times (25 \times 10^{-4})^{3/2}}$$

$$B = 36\pi \times 10^{-7}\ \hat{k}$$

Final Answer

$B = 36\pi \times 10^{-7}$ T $\approx 1.13 \times 10^{-5}$ T, directed along the $+Z$-axis.

Common Mistake: Remember to convert the radius and axial distance from cm to metres before substituting into SI formulas – mixing units is the most common source of error in this problem.

II. Magnetic Force on Moving Charge

Question 14

An electron, moving with a velocity of 5×10^7 m s^{-1}, enters a magnetic field of 1 Wb m^{-2} at an angle of 30°. Calculate the force on the electron.

Given: $v = 5 \times 10^7$ m s^{-1}, $B = 1$ T, $\theta = 30°$

Formula: $F = qvB\sin\theta$

Solution:

$$F = 1.6 \times 10^{-19} \times 5 \times 10^7 \times 1 \times \sin 30° = 1.6 \times 10^{-19} \times 5 \times 10^7 \times 0.5$$

$$F = 4.0 \times 10^{-12} \text{ N}$$

Final Answer

$F = 4.0 \times 10^{-12}$ N

Question 15

A 2 MeV proton is moving perpendicular to a uniform magnetic field of 2.5 T. Find the force on the proton. (Take mass of proton as 1.65×10^{-27} kg.)

Given: $KE = 2$ MeV $= 3.2 \times 10^{-13}$ J, $B = 2.5$ T, $m_p = 1.65 \times 10^{-27}$ kg

Formula: $v = \sqrt{\frac{2\,KE}{m}}$; $F = qvB$ (perpendicular, $\sin\theta = 1$)

Solution:

$$v = \sqrt{\frac{2 \times 3.2 \times 10^{-13}}{1.65 \times 10^{-27}}} = 1.97 \times 10^7 \text{ m s}^{-1}$$

$$F = 1.6 \times 10^{-19} \times 1.97 \times 10^7 \times 2.5 = 7.88 \times 10^{-12} \text{ N}$$

Final Answer

$F = 7.88 \times 10^{-12}$ N

Common Mistake: Convert MeV to Joules (1 eV $= 1.6 \times 10^{-19}$ J) before using it in the kinetic-energy formula – using MeV directly is a common slip.

Question 16

An electron is moving vertically upwards with a speed of 2.0×10^8 m s^{-1}. What will be the magnitude and direction of the force on the electron exerted by a horizontal magnetic field of 0.50 N A^{-1}m^{-1} directed towards west? What will be the acceleration of the electron?

★ **Likely Exam Question**

Given: $v = 2.0 \times 10^8$ m s^{-1} (vertical), $B = 0.50$ T (horizontal, west), $m_e = 9.1 \times 10^{-31}$ kg

Formula: $F = qvB\sin\theta$ ($\theta = 90°$ since $v \perp B$); $a = \dfrac{F}{m}$

Solution:

$$F = 1.6 \times 10^{-19} \times 2.0 \times 10^8 \times 0.50 = 1.6 \times 10^{-11} \text{ N}$$

Direction: for a positive charge, $\vec{v} \times \vec{B}$ (up × west) points south; since the electron carries negative charge, the force is reversed – it acts towards *north*.

$$a = \frac{1.6 \times 10^{-11}}{9.1 \times 10^{-31}} = 1.8 \times 10^{19} \text{ m s}^{-2}$$

Final Answer

$F = 1.6 \times 10^{-11}$ N, towards north; $a = 1.8 \times 10^{19}$ m s^{-2}

Common Mistake: Forgetting to reverse the direction obtained from $\vec{v} \times \vec{B}$ because the electron's charge is negative is the single most common mistake in this type of problem.

Question 17

An electron accelerated by a potential difference of 300 V is moving at a distance of 4 mm from a long, straight wire. If a current of 5 A flows through the wire, what is the force acting on the electron?

Given: $V = 300$ V, $r = 4$ mm $= 4 \times 10^{-3}$ m, $I = 5$ A

Formula: $v = \sqrt{\dfrac{2eV}{m_e}}$; $B = \dfrac{\mu_0 I}{2\pi r}$; $F = evB$ (electron moves parallel to wire, so $v \perp B$)

Solution:

$$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 300}{9.1 \times 10^{-31}}} = 1.03 \times 10^{7} \text{ m s}^{-1}$$

$$B = \frac{2 \times 10^{-7} \times 5}{4 \times 10^{-3}} = 2.5 \times 10^{-4} \text{ T}$$

$$F = 1.6 \times 10^{-19} \times 1.03 \times 10^{7} \times 2.5 \times 10^{-4} = 4.0 \times 10^{-16} \text{ N}$$

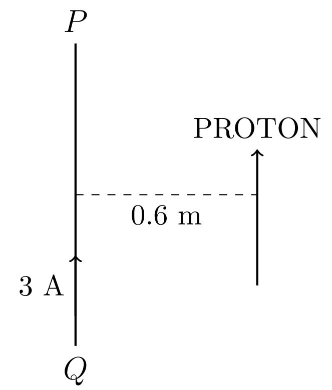
Final Answer

$F = 4 \times 10^{-16}$ N

Question 18

PQ is a long straight conductor carrying a current of 3 A, as shown. A proton moves with a velocity of 2×10^{7} m s^{-1} parallel to it, at a distance of 0.6 m from PQ. Find the force acting on the proton. *(ISC 2017)*

★ Likely Exam Question



PQ carries current 3 A; a proton moves parallel to it at 0.6 m with velocity v

Given: $I = 3$ A, $v = 2 \times 10^{7}$ m s^{-1}, $r = 0.6$ m

Formula: $B = \dfrac{\mu_0 I}{2\pi r}$; $F = qvB$

Solution:

$$B = \frac{2 \times 10^{-7} \times 3}{0.6} = 1.0 \times 10^{-6} \text{ T}$$

$$F = 1.6 \times 10^{-19} \times 2 \times 10^{7} \times 1.0 \times 10^{-6} = 3.2 \times 10^{-18} \text{ N}$$

Final Answer

$F = 3.2 \times 10^{-18}$ N, towards the right, away from the wire.

Common Mistake: This is a repeated ISC board question – practise the direction reasoning carefully using the right-hand rule for both the field due to the wire and the force on the moving charge.

III. Path of Charged Particle in Uniform Magnetic Field

Question 19

A uniform magnetic field of 6.5 G (1 G = 10^{-4} T) is maintained in a chamber. An electron is shot into the field with a speed of 4.8×10^6 m s^{-1} normal to the field. Determine the radius of the circular orbit and the frequency of revolution of the electron. Does the frequency depend on the speed of the electron?

★ Likely Exam Question

Given: $B = 6.5 \text{ G} = 6.5 \times 10^{-4}$ T, $v = 4.8 \times 10^6$ m s^{-1} (normal to field)

Formula: $r = \dfrac{m_e v}{eB}$; $f = \dfrac{eB}{2\pi m_e}$

Solution:

$$r = \frac{9.1 \times 10^{-31} \times 4.8 \times 10^6}{1.6 \times 10^{-19} \times 6.5 \times 10^{-4}} = 4.2 \times 10^{-2} \text{ m} = 4.2 \text{ cm}$$

$$f = \frac{1.6 \times 10^{-19} \times 6.5 \times 10^{-4}}{2\pi \times 9.1 \times 10^{-31}} = 1.8 \times 10^7 \text{ Hz} = 18 \text{ MHz}$$

Since $f = \dfrac{eB}{2\pi m}$ has no v term, the frequency does *not* depend on the electron's speed.

Final Answer

$r = 4.2$ cm, $f = 18$ MHz, frequency is independent of speed.

Question 20

An α-particle is moving along a circle of radius 0.45 m in a magnetic field of 12 T. Find (i) the speed of the particle, (ii) the time-period of the particle. (Mass of α-particle is 6.7×10^{-27} kg.)

Given: $r = 0.45$ m, $B = 12$ T, $q = 2e = 3.2 \times 10^{-19}$ C, $m_\alpha = 6.7 \times 10^{-27}$ kg

Formula: $v = \dfrac{qBr}{m}$; $T = \dfrac{2\pi m}{qB}$

Solution:

$$v = \frac{3.2 \times 10^{-19} \times 12 \times 0.45}{6.7 \times 10^{-27}} = 2.58 \times 10^{8} \text{ m s}^{-1}$$
$$T = \frac{2\pi \times 6.7 \times 10^{-27}}{3.2 \times 10^{-19} \times 12} = 1.09 \times 10^{-8} \text{ s}$$

Final Answer

(i) $v = 2.58 \times 10^{8}$ m s^{-1} (ii) $T = 1.09 \times 10^{-8}$ s

Question 21

An electron of 40 eV energy is revolving in a circular path in a magnetic field of 9×10^{-5} T. Determine: (i) speed of the electron, (ii) radius of the circular path.

Given: $KE = 40 \text{ eV} = 6.4 \times 10^{-18}$ J, $B = 9 \times 10^{-5}$ T

Formula: $v = \sqrt{\dfrac{2\,KE}{m_e}}$; $r = \dfrac{m_e v}{eB}$

Solution:

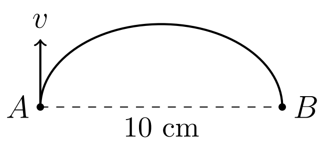
$$v = \sqrt{\frac{2 \times 6.4 \times 10^{-18}}{9.1 \times 10^{-31}}} = 3.75 \times 10^{6} \text{ m s}^{-1}$$
$$r = \frac{9.1 \times 10^{-31} \times 3.75 \times 10^{6}}{1.6 \times 10^{-19} \times 9 \times 10^{-5}} = 0.237 \text{ m} = 23.7 \text{ cm}$$

Final Answer

(i) $v = 3.75 \times 10^{6}$ m s^{-1} (ii) $r = 23.7$ cm

Question 22

The velocity of an electron at point A is 10^{7} m s^{-1}. Find: (i) the direction and magnitude of the magnetic field due to which the electron will go from A to B on the semi-circular path shown (AB = 10 cm), (ii) the time taken by the electron in going from A to B.



Electron travels from A to B along the semicircular path shown (AB = 10 cm)

Given: $v = 10^{7}$ m s^{-1}, diameter $AB = 10 \text{ cm} \Rightarrow r = 5 \text{ cm} = 0.05$ m

Formula: $r = \dfrac{m_e v}{eB} \Rightarrow B = \dfrac{m_e v}{er}$; $t = \dfrac{\pi r}{v}$ (time for a half-circle)

Solution:

$$B = \frac{9.1 \times 10^{-31} \times 10^{7}}{1.6 \times 10^{-19} \times 0.05} = 1.12 \times 10^{-3}\ \text{T}$$
$$t = \frac{\pi \times 0.05}{10^{7}} = 1.57 \times 10^{-8}\ \text{s}$$

Final Answer

(i) $B = 1.12 \times 10^{-3}$ T, perpendicular to the plane of paper, pointing downwards
(ii) $t = 1.57 \times 10^{-8}$ s

Question 23

An electron is moving with a speed of 3.0×10^{7} m s^{-1} along a circle in a magnetic field of 0.50 N A^{-1}m^{-1}. Calculate the radius of the circle and the kinetic energy of the electron.

Given: $v = 3.0 \times 10^{7}$ m s^{-1}, $B = 0.50$ T

Formula: $r = \dfrac{m_e v}{eB}$; $KE = \frac{1}{2} m_e v^2$

Solution:

$$r = \frac{9.1 \times 10^{-31} \times 3.0 \times 10^{7}}{1.6 \times 10^{-19} \times 0.50} = 3.4 \times 10^{-4}\ \text{m} = 0.34\ \text{mm}$$
$$KE = \frac{1}{2} \times 9.1 \times 10^{-31} \times (3.0 \times 10^{7})^2 = 4.1 \times 10^{-16}\ \text{J}$$

Final Answer

$r = 0.34$ mm, $KE = 4.1 \times 10^{-16}$ J

Question 24

An electron emitted by a heated cathode and accelerated through a potential difference of 2 kV, enters a region with a uniform magnetic field of 0.15 T. Determine the trajectory and radius of the electron's path if the field is (a) transverse to its initial velocity, (b) makes an angle of 30° with the initial velocity.

★ **Likely Exam Question**

Given: $V = 2$ kV $= 2000$ V, $B = 0.15$ T

Formula: $KE = eV = \frac{1}{2} m_e v^2 \Rightarrow v = \sqrt{\dfrac{2eV}{m_e}}$; $r = \dfrac{m_e v}{eB\sin\theta}$

Solution:

$$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 2000}{9.1 \times 10^{-31}}} = 2.65 \times 10^{7} \text{ m s}^{-1}$$

(a) When B is transverse to $\vec{v}$ ($\theta = 90^\circ$), the path is a *circle*:

$$r = \frac{m_e v}{eB} = \frac{9.1 \times 10^{-31} \times 2.65 \times 10^{7}}{1.6 \times 10^{-19} \times 0.15} = 9.95 \times 10^{-4} \text{ m}$$

(b) When B makes 30° with $\vec{v}$, the velocity component perpendicular to B is $v \sin 30^\circ$; the path becomes a *helix (spiral)* of radius:

$$r' = \frac{m_e v}{eB \sin 30^\circ} = \frac{r}{\sin 30^\circ} = 2r = 2 \times 9.95 \times 10^{-4} = 1.99 \times 10^{-3} \text{ m}$$

Final Answer

(a) Circular path, $r = 9.95 \times 10^{-4}$ m (b) Helical (spiral) path, $r' = 2r = 1.99 \times 10^{-3}$ m

Common Mistake: When the field is oblique to the velocity, the path is a helix, not a simple circle – only the velocity component perpendicular to B determines the radius; the parallel component causes uniform drift along the field direction.

Question 25

A proton, a deuteron and an α-particle, whose kinetic energies are the same, enter perpendicularly into a magnetic field. Compare the radii of their circular paths.

★ **Likely Exam Question**

Given: KE same for all three; proton: mass m, charge e; deuteron: mass $2m$, charge e; α-particle: mass $4m$, charge $2e$

Formula: $r = \dfrac{mv}{qB}$ with $v = \sqrt{\dfrac{2\,KE}{m}} \Rightarrow r = \dfrac{\sqrt{2m \cdot KE}}{qB}$, i.e. $r \propto \dfrac{\sqrt{m}}{q}$

Solution:

$$r_p \propto \frac{\sqrt{m}}{e}, \qquad r_d \propto \frac{\sqrt{2m}}{e} = \sqrt{2}\, r_p, \qquad r_\alpha \propto \frac{\sqrt{4m}}{2e} = \frac{2\sqrt{m}}{2e} = r_p$$

Final Answer

$r_d = \sqrt{2}\, r_p, \qquad r_\alpha = r_p, \qquad r_d = \sqrt{2}\, r_\alpha$

Common Mistake: It is tempting to assume the heaviest particle (α) has the largest radius – but its doubled charge exactly cancels the extra mass, leaving $r_\alpha = r_p$, the same as the proton.

IV. Magnetic Force on Current Carrying Conductor

Question 26

Find the force per unit length on a wire carrying a current of 8 A and making a 30° angle with the direction of a uniform magnetic field of 0.15 T.

★ **Likely Exam Question**

Given: $I = 8$ A, $\theta = 30°$, $B = 0.15$ T

Formula: $\dfrac{F}{L} = BI\sin\theta$

Solution:

$$\frac{F}{L} = 0.15 \times 8 \times \sin 30° = 0.15 \times 8 \times 0.5 = 0.6 \text{ N m}^{-1}$$

Final Answer

$F/L = 0.6$ N m^{-1}

Question 27

A current of 5.0 A is flowing upwards in a long vertical wire placed in a uniform horizontal northward magnetic field of 0.020 T. How much force and in what direction will the field exert on 0.06 m length of the wire?

Given: $I = 5.0$ A (vertical), $B = 0.020$ T (horizontal, north), $L = 0.06$ m

Formula: $F = BIL\sin\theta$ ($\theta = 90°$ since current is vertical and B is horizontal)

Solution:

$$F = 0.020 \times 5.0 \times 0.06 = 6.0 \times 10^{-3} \text{ N}$$

Direction: current is up, B is north; by $\vec{F} = I\vec{L} \times \vec{B}$, the force acts towards the *west*.

Final Answer

$F = 6 \times 10^{-3}$ N, towards west.

Question 28

A current-carrying conductor has 8.0×10^{22} free electrons per metre length and their mean drift velocity is 8.0×10^{-5} m s^{-1}. If a magnetic field of 0.10 T is applied perpendicular to the conductor, calculate (i) current in the conductor, (ii) force on an electron, (iii) force imposed per metre length of the conductor.

Given: $nA = 8.0 \times 10^{22}$ m^{-1}, $v_d = 8.0 \times 10^{-5}$ m s^{-1}, $B = 0.10$ T

Formula: $I = nAv_de$; $F_{electron} = ev_dB$; $F/L = BIL/L = BI$ (per unit length)

Solution:

(i)

$$I = 8.0 \times 10^{22} \times 8.0 \times 10^{-5} \times 1.6 \times 10^{-19} = 1.02 \text{ A}$$

(ii)

$$F_{electron} = 1.6 \times 10^{-19} \times 8.0 \times 10^{-5} \times 0.10 = 1.28 \times 10^{-24} \text{ N}$$

(iii) For 1 m length:

$$F = BIL = 0.10 \times 1.02 \times 1 = 0.1024 \text{ N}$$

Final Answer

(i) $I = 1.02$ A (ii) $F_{electron} = 1.28 \times 10^{-24}$ N (iii) $F = 0.1024$ N per metre

Common Mistake: Do not multiply nA (already given as electrons per unit length) by an extra length factor – the quantity nA already has the length built into it, per the hint $I = nAv_de$.

Question 29

Two parallel rails, distant 2.0×10^{-2} m apart, are laid in a north-south direction. On these rails is kept a metal cylinder of mass 4.0×10^{-2} kg. A battery is connected to the rails with its positive terminal joined to the rail on the eastern side. The battery sends a current of 3.0 A in the cylinder. If a uniform magnetic field of 1.2 T directed upwards is applied there, find the magnitude and direction of the magnetic force imposed upon the cylinder. What would be the acceleration in the cylinder?

Given: $L = 2.0 \times 10^{-2}$ m, $m = 4.0 \times 10^{-2}$ kg, $I = 3.0$ A, $B = 1.2$ T (vertical)

Formula: $F = BIL$ (current across rails is perpendicular to vertical field); $a = F/m$

Solution:

$$F = 1.2 \times 3.0 \times 2.0 \times 10^{-2} = 7.2 \times 10^{-2} \text{ N}$$

$$a = \frac{7.2 \times 10^{-2}}{4.0 \times 10^{-2}} = 1.8 \text{ m s}^{-2}$$

Direction: by $\vec{F} = I\vec{L} \times \vec{B}$ with current flowing east-to-west across the cylinder and B vertically upward, the force acts along the rails towards the *north*.

Final Answer

$F = 7.2 \times 10^{-2}$ N, towards north; $a = 1.8$ m s^{-2}

Common Mistake: Since the current here flows *across* the rails (not along them), and B is vertical, the force direction ends up *along* the rails – students often get confused about which direction is "current direction" in this rail-cylinder setup.

Question 30

On a 30° inclined smooth plane, a thin current-carrying rod is placed parallel to the horizontal ground. The plane is located in a uniform vertical magnetic field of 0.15 T. The mass/length of the rod is 0.30 kg m^{-1}. Find the current required to keep the rod stationary.

Given: $\theta = 30°$, $B = 0.15$ T, mass/length $= 0.30$ kg m^{-1}

Formula: For equilibrium: $BIL\cos\theta = mg\sin\theta$ (per unit length: $BI\cos\theta = (m/L)g\sin\theta$)

Solution:

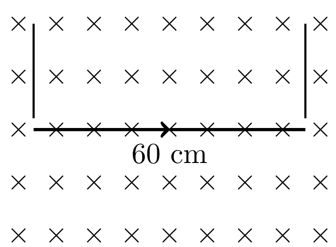
$$I = \frac{(m/L)\,g\tan\theta}{B} = \frac{0.30 \times 9.8 \times \tan 30°}{0.15} = \frac{0.30 \times 9.8 \times 0.577}{0.15}$$
$$I = 11.3 \text{ A}$$

Final Answer

$I = 11.3$ A

Question 31

A 60 cm long wire (mass 10 g) is hung by two flexible wires in a magnetic field of 0.40 T. Find the magnitude and direction of the current required to neutralize the tension in the hanging wires.



60 cm wire hung by two flexible wires, in a magnetic field directed into the page

Given: $L = 60$ cm $= 0.6$ m, $m = 10$ g $= 0.01$ kg, $B = 0.40$ T

Formula: For the tension to vanish, the magnetic force must support the weight: $BIL = mg$

Solution:

$$I = \frac{mg}{BL} = \frac{0.01 \times 9.8}{0.40 \times 0.6} = \frac{0.098}{0.24} = 0.41 \text{ A}$$

Final Answer

$I = 0.41$ A, flowing from left to right.

V. Force between Parallel Straight Current-Carrying Conductors

Question 32

Two long, parallel, straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are placed 4.0 cm apart. Estimate the force on a 10 cm section of wire A.

★ **Likely Exam Question**

Given: $I_1 = 8.0$ A, $I_2 = 5.0$ A, $d = 4.0$ cm $= 0.04$ m, $L = 10$ cm $= 0.1$ m

Formula: $\dfrac{F}{L} = \dfrac{\mu_0 I_1 I_2}{2\pi d}$

Solution:

$$\frac{F}{L} = \frac{2 \times 10^{-7} \times 8.0 \times 5.0}{0.04} = 2 \times 10^{-4} \text{ N m}^{-1}$$
$$F = 2 \times 10^{-4} \times 0.1 = 2 \times 10^{-5} \text{ N}$$

Final Answer

$F = 2 \times 10^{-5}$ N, towards B (attractive, since currents are in the same direction).

Question 33

A long horizontal rigidly-supported wire carries a current of 100 A. Directly above it and parallel to it is a fine wire that carries a current of 200 A and weighs 0.05 N/m. How far above the lower wire should the fine wire be kept to support it by magnetic repulsion?

Given: $I_1 = 100$ A, $I_2 = 200$ A, weight/length $= 0.05$ N m^{-1}

Formula: For equilibrium, magnetic force per length = weight per length: $\dfrac{\mu_0 I_1 I_2}{2\pi d} = \dfrac{mg}{L}$

Solution:

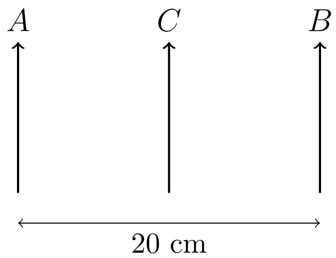
$$d = \frac{\mu_0 I_1 I_2}{2\pi \times (mg/L)} = \frac{2 \times 10^{-7} \times 100 \times 200}{0.05} = \frac{4 \times 10^{-3}}{0.05} = 0.08 \text{ m}$$

Final Answer

$d = 8$ cm

Question 34

Two very long, straight, parallel wires A and B carry currents of 10 A and 20 A respectively and are at a distance 20 cm apart. If a third wire C (length 15 cm) having a current of 10 A is placed midway between them, how much force will act on C? The direction of current in all three wires is the same.



Wires A, B carry 10 A and 20 A; wire C (10 A) is placed midway between them

Given: $I_A = 10$ A, $I_B = 20$ A, $I_C = 10$ A, $d = 20$ cm, so C is 10 cm from each wire, $L_C = 15$ cm $= 0.15$ m

Formula: $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$, applied separately for A-C and B-C (forces act in opposite senses since A, B are on opposite sides of C)

Solution:

Force per length on C due to A (attractive, towards A):

$$\left(\frac{F}{L}\right)_A = \frac{2 \times 10^{-7} \times 10 \times 10}{0.1} = 2 \times 10^{-4} \text{ N m}^{-1}$$

Force per length on C due to B (attractive, towards B):

$$\left(\frac{F}{L}\right)_B = \frac{2 \times 10^{-7} \times 20 \times 10}{0.1} = 4 \times 10^{-4} \text{ N m}^{-1}$$

Since these pull C in opposite directions, net force per length $= 4 \times 10^{-4} - 2 \times 10^{-4} = 2 \times 10^{-4}$ N m^{-1}, directed towards B.

$$F = 2 \times 10^{-4} \times 0.15 = 3.0 \times 10^{-5} \text{ N}$$

Final Answer

$F = 3.0 \times 10^{-5}$ N, towards B.

Common Mistake: Because C sits between A and B, the two attractive forces act in *opposite* directions on C – they must be subtracted, not added.

VI. Cyclotron

Question 35

Calculate the magnetic field in which the cyclotron dees should be placed to accelerate protons. The applied frequency is 8 megacycles/second. Take proton mass $= 1.67 \times 10^{-27}$ kg.

★ **Likely Exam Question**

Given: $f = 8$ MHz $= 8 \times 10^6$ Hz, $\quad m_p = 1.67 \times 10^{-27}$ kg

Formula: $B = \dfrac{2\pi m f}{q}$

Solution:

$$B = \frac{2\pi \times 1.67 \times 10^{-27} \times 8 \times 10^6}{1.6 \times 10^{-19}} = 0.524 \text{ T}$$

Final Answer

$B = 0.524$ T

Question 36

A cyclotron accelerating protons has dees of 0.4 m radius and a radio-frequency supply of 12 megacycles/second at 5000 V maximum value. Calculate the magnitude of the required magnetic field, the energy and velocity acquired by the protons. Take proton mass $= 1.67 \times 10^{-27}$ kg.

Given: $R = 0.4$ m, $\quad f = 12$ MHz, $\quad m_p = 1.67 \times 10^{-27}$ kg

Formula: $B = \dfrac{2\pi m f}{q}$; $\qquad v_{max} = 2\pi f R$; $\qquad E_{max} = \frac{1}{2} m v_{max}^2$

Solution:

$$B = \frac{2\pi \times 1.67 \times 10^{-27} \times 12 \times 10^6}{1.6 \times 10^{-19}} = 0.786 \text{ T}$$

$$v_{max} = 2\pi \times 12 \times 10^6 \times 0.4 = 3.0 \times 10^7 \text{ m s}^{-1}$$

$$E_{max} = \frac{1}{2} \times 1.67 \times 10^{-27} \times (3.0 \times 10^7)^2 = 7.6 \times 10^{-13} \text{ J} = 4.73 \text{ MeV}$$

Final Answer

$B = 0.786$ T, $\quad E_{max} = 4.73$ MeV, $\quad v_{max} = 3 \times 10^7$ m s^{-1}

Common Mistake: The 5000 V accelerating voltage given in the question is a distractor for finding B; it is not needed unless verifying the number of accelerations – speed and energy come directly from the geometry ($v = 2\pi f R$).

VII. Torque on Current Carrying Coil in Magnetic Field

Question 37

When a coil of magnetic moment 2.5×10^{-8} J T^{-1} is placed with its face parallel to a uniform magnetic field, it experiences a torque of 7.5×10^{-9} J. Find the magnitude of the field.

Given: $m = 2.5 \times 10^{-8}$ J T^{-1}, $\tau = 7.5 \times 10^{-9}$ J

Formula: With the coil's face parallel to B, the normal is perpendicular to B ($\theta = 90°$) – this is the condition for *maximum* torque: $\tau_{max} = mB$

Solution:

$$B = \frac{\tau}{m} = \frac{7.5 \times 10^{-9}}{2.5 \times 10^{-8}} = 0.30 \text{ T}$$

Final Answer

$B = 0.30$ T

Common Mistake: "Face parallel to field" means the coil's *normal* is perpendicular to B – this gives *maximum* torque, not zero torque. Confusing "face parallel" with "normal parallel" is a very common conceptual trap.

Question 38

A square coil of side 10 cm consists of 20 turns and carries a current of 12 A. It is suspended vertically in a uniform horizontal magnetic field of 0.80 T, such that the normal to the plane of the coil makes an angle of 30° with the field direction. Find the magnitude of the torque experienced by the coil.

Given: $a = 10$ cm $= 0.1$ m, $N = 20$, $I = 12$ A, $B = 0.80$ T, $\theta = 30°$ (angle of normal with field)

Formula: $\tau = NBIA \sin\theta$

Solution:

$$A = (0.1)^2 = 0.01 \text{ m}^2$$
$$\tau = 20 \times 0.80 \times 12 \times 0.01 \times \sin 30° = 20 \times 0.80 \times 12 \times 0.01 \times 0.5 = 0.96 \text{ N m}$$

Final Answer

$\tau = 0.96$ N m

Question 39

A rectangular loop of area 5 m^2, has 50 turns and carries a current of 1 A. It is held in a uniform magnetic field of 0.1 T, at an angle of 30° (with the plane of the coil). Calculate the torque experienced by the coil. *(ISC 2019)*

★ Likely Exam Question

Given: $A = 5\text{ m}^2$, $N = 50$, $I = 1\text{ A}$, $B = 0.1\text{ T}$, angle between B and *plane* of coil $= 30°$

Formula: $\tau = NBIA\sin\theta$, where θ is the angle between the normal and B. Since 30° is given with the *plane*, the angle with the normal is $90° - 30° = 60°$.

Solution:

$$\tau = 50 \times 0.1 \times 1 \times 5 \times \sin 60° = 50 \times 0.1 \times 1 \times 5 \times 0.866$$

$$\tau = 21.65\text{ N m}$$

Final Answer

$\tau = 21.65\text{ N m}$

Common Mistake: This is a classic ISC trap: when the angle is stated with respect to the *plane* of the coil, you must use its complement with the *normal* in the torque formula $\tau = NBIA\sin\theta$. Using $\sin 30°$ directly gives a wrong answer.

Question 40

A circular coil of radius 6.0 cm and 25 turns carries a current of 10 A. It is suspended vertically in a uniform magnetic field of 1.2 T and the field lines are horizontal, in the plane of the coil. Compute the torque acting on the coil.

Given: $r = 6.0\text{ cm} = 0.06\text{ m}$, $N = 25$, $I = 10\text{ A}$, $B = 1.2\text{ T}$, field lines lie *in the plane* of the coil

Formula: $\tau = NBIA\sin\theta$; since B lies in the plane of the coil, the normal is perpendicular to B ($\theta = 90°$, maximum torque condition)

Solution:

$$A = \pi r^2 = \pi \times (0.06)^2 = 0.01131\text{ m}^2$$

$$\tau = 25 \times 1.2 \times 10 \times 0.01131 = 3.4\text{ N m}$$

Final Answer

$\tau = 3.4\text{ N m}$

Question 41

A coil of area 5 cm^2 and having 100 turns is placed in a uniform magnetic field of 1.5 T. When a current of 0.2 A is passed through the coil, find the magnetic dipole moment of the coil and the maximum torque produced.

Given: $A = 5 \text{ cm}^2 = 5 \times 10^{-4} \text{ m}^2$, $N = 100$, $B = 1.5$ T, $I = 0.2$ A

Formula: $m = NIA$; $\tau_{max} = mB$

Solution:

$$m = 100 \times 0.2 \times 5 \times 10^{-4} = 1.0 \times 10^{-2} \text{ A m}^2$$
$$\tau_{max} = 1.0 \times 10^{-2} \times 1.5 = 1.5 \times 10^{-2} \text{ N m}$$

Final Answer

$m = 1.0 \times 10^{-2}$ A m^2, $\tau_{max} = 1.5 \times 10^{-2}$ N m

Question 42

A rectangular coil of sides 8.0 cm and 6.0 cm having 2000 turns and carrying a current of 0.20 A is suspended in a uniform magnetic field of 0.20 T directed along the positive X-axis. (a) What is the net force on the coil? (b) What is the maximum torque the coil can experience, and in which orientation? (c) For which orientation of the coil is the torque zero? When is this equilibrium stable and when is it unstable?

★ **Likely Exam Question**

Given: $a = 8.0 \text{ cm} \times 6.0 \text{ cm} = 0.08 \times 0.06 \text{ m}^2$, $N = 2000$, $I = 0.20$ A, $B = 0.20$ T

Formula: Net force on any closed current loop in a uniform field = 0; $\tau_{max} = NBIA$

Solution:

(a) The net force on a closed current loop in a *uniform* magnetic field is always zero, regardless of orientation.

$$F_{net} = 0$$

(b)

$$A = 0.08 \times 0.06 = 4.8 \times 10^{-3} \text{ m}^2$$
$$\tau_{max} = NBIA = 2000 \times 0.20 \times 0.20 \times 4.8 \times 10^{-3} = 0.384 \text{ N m}$$

This maximum occurs when the plane of the coil is *parallel* to B (normal perpendicular to B).

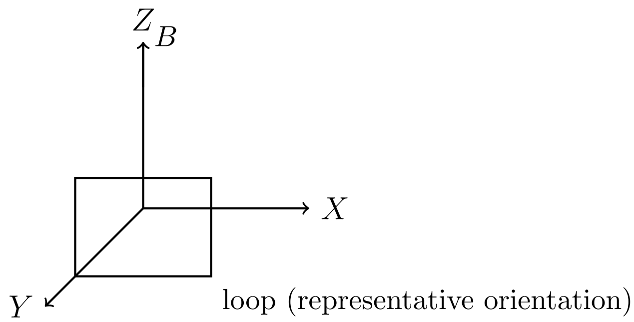
(c) Torque is zero when the plane of the coil is *perpendicular* to B, i.e. the normal to the coil is parallel (or antiparallel) to B. The equilibrium is *stable* when the magnetic moment $\vec{m}$ is parallel to $\vec{B}$, and *unstable* when $\vec{m}$ is antiparallel to $\vec{B}$.

Final Answer

(a) $F_{net} = 0$ (b) $\tau_{max} = 0.384$ N m (plane parallel to B) (c) $\tau = 0$ when normal is along B; stable if $\vec{m} \parallel \vec{B}$, unstable if $\vec{m}$ antiparallel to $\vec{B}$.

Question 43

A uniform magnetic field of 3000 gauss is established along the positive Z-axis. A rectangular loop of size 10 cm $\times$ 5 cm carries a current of 12 A. What is the magnitude and direction of the torque on the loop in the four orientations shown in the figure?



Rectangular loop in field B (along $+Z$) shown in orientations (a)-(d) – normal along Y, along Y (reversed sense), along X, and along Z

Given: $B = 3000$ G $= 0.3$ T (along $+Z$), $A = 0.10 \times 0.05 = 5 \times 10^{-3}$ m^2, $I = 12$ A

Formula: $m = NIA$ (here $N = 1$); $\tau = mB\sin\theta$, θ = angle between coil's normal and B

Solution:

$$m = 1 \times 12 \times 5 \times 10^{-3} = 0.06 \text{ A m}^2$$

(a) Coil's normal along Y-axis, B along Z-axis: $\theta = 90^\circ$

$$\tau = 0.06 \times 0.3 = 1.8 \times 10^{-2} \text{ N m, along} - Y$$

(b) Same geometry as (a) (current sense in the loop's plane changed but normal still along Y): $\tau = 1.8 \times 10^{-2}$ N m, along $-Y$

(c) Coil's normal along X-axis, B along Z-axis: $\theta = 90^\circ$

$$\tau = 1.8 \times 10^{-2} \text{ N m, along} - X$$

(d) Coil's normal along Z-axis, parallel to B: $\theta = 0^\circ \Rightarrow \tau = 0$

Final Answer

(a) and (b) $\tau = 1.8 \times 10^{-2}$ N m, along $-Y$ (c) $\tau = 1.8 \times 10^{-2}$ N m, along $-X$ (d) $\tau = 0$

Common Mistake: Always convert gauss to tesla (1 G $= 10^{-4}$ T) before substituting – forgetting this factor of 10^{-4} is one of the most common numerical slips in magnetism problems.

VIII. Moving Coil Galvanometer and Voltage/Current Sensitivities

Question 44

A moving-coil galvanometer has its coil of 24 turns and 12 Ω resistance. By how much should the number of turns be increased so that the current-sensitivity of the galvanometer increases by 25%? If, in doing so, the resistance of the coil increases to 20 Ω, what will be the effect on the voltage-sensitivity of the galvanometer?

Given: $N_1 = 24, \quad R_1 = 12\ \Omega; \quad \text{new } R_2 = 20\ \Omega$; current sensitivity to increase by 25%

Formula: $I_s = \dfrac{NBA}{k} \propto N$ (for fixed B, A, k); $\qquad V_s = \dfrac{I_s}{R}$

Solution:

Since $I_s \propto N$, a 25% increase in I_s requires:

$$N_2 = 1.25 \times N_1 = 1.25 \times 24 = 30$$

Voltage sensitivity ratio:

$$\frac{V_{s2}}{V_{s1}} = \frac{I_{s2}/R_2}{I_{s1}/R_1} = \frac{1.25\, I_{s1}}{I_{s1}} \times \frac{R_1}{R_2} = 1.25 \times \frac{12}{20} = 0.75$$

Final Answer

Turns must be increased from 24 to 30; voltage-sensitivity will fall to 0.75 of its original value (a 25% decrease).

Common Mistake: Increasing the number of turns raises current sensitivity but simultaneously raises coil resistance – students often forget that voltage sensitivity depends on *both* and can actually decrease even though current sensitivity increases.

Question 45

To increase the current-sensitivity of a moving-coil galvanometer by 50%, its resistance is increased so that the new resistance becomes twice its initial resistance. By what factor does its voltage-sensitivity change?

Given: $I_{s2} = 1.5\, I_{s1}, \quad R_2 = 2R_1$

Formula: $V_s = \dfrac{I_s}{R}$

Solution:

$$\frac{V_{s2}}{V_{s1}} = \frac{I_{s2}/R_2}{I_{s1}/R_1} = \frac{1.5\,I_{s1}}{I_{s1}} \times \frac{R_1}{2R_1} = 1.5 \times 0.5 = 0.75$$

Final Answer

Voltage-sensitivity is reduced to 0.75 times (75%) of its previous value.

IX. Conversion of Galvanometer into Ammeter/Voltmeter

Question 46

A galvanometer has a resistance of 100 Ω. A PD of 100 mV between its terminals gives a full-scale deflection. Calculate the shunt resistance required to convert the galvanometer into an ammeter reading up to 5 A.

★ **Likely Exam Question**

Given: $R_g = 100\ \Omega$, full-scale PD $= 100$ mV $= 0.1$ V, desired range $I = 5$ A

Formula: $I_g = \dfrac{V}{R_g}$; $S = \dfrac{I_g R_g}{I - I_g}$

Solution:

$$I_g = \frac{0.1}{100} = 1 \times 10^{-3}\ \text{A}$$

$$S = \frac{1 \times 10^{-3} \times 100}{5 - 0.001} = \frac{0.1}{4.999} \approx 0.02\ \Omega$$

Final Answer

$S = 0.02\ \Omega$

Question 47

A galvanometer of resistance 99 Ω gives a full-scale deflection for a current of 50 mA. Find the value of the shunt required to convert it into an ammeter reading up to a maximum current of 5 A. What is the resistance of the ammeter?

Given: $R_g = 99\ \Omega$, $I_g = 50$ mA $= 0.05$ A, $I = 5$ A

Formula: $S = \dfrac{I_g R_g}{I - I_g}$; $R_{ammeter} = \dfrac{R_g S}{R_g + S}$

Solution:

$$S = \frac{0.05 \times 99}{5 - 0.05} = \frac{4.95}{4.95} = 1.0\ \Omega$$

$$R_{ammeter} = \frac{99 \times 1.0}{99 + 1.0} = \frac{99}{100} = 0.99\ \Omega$$

Final Answer

$S = 1.0\ \Omega, \quad R_{ammeter} = 0.99\ \Omega$

Question 48

The coil of a galvanometer has a resistance of 100 Ω. It shows full-scale deflection for a current of 5×10^{-4} A. How will you convert it into a voltmeter to read a maximum PD of 5 V?

Given: $R_g = 100\ \Omega, \quad I_g = 5 \times 10^{-4}\ \text{A}, \quad V = 5\ \text{V}$

Formula: $R = \dfrac{V}{I_g} - R_g$

Solution:

$$R = \frac{5}{5 \times 10^{-4}} - 100 = 10000 - 100 = 9900\ \Omega$$

Final Answer

Add a resistance of 9900 Ω in series with the galvanometer.

Question 49

The resistance of a voltmeter is 180 Ω. When there is a 5 V potential difference across its ends, it gives full-scale deflection. What resistance should be connected in series to obtain full-scale deflection at 25 V?

Given: $R_v = 180\ \Omega$, full-scale at $V_1 = 5$ V; extend range to $V_2 = 25$ V

Formula: $I_g = \dfrac{V_1}{R_v}; \qquad R = \dfrac{V_2}{I_g} - R_v$

Solution:

$$I_g = \frac{5}{180} = 0.02778\ \text{A}$$
$$R = \frac{25}{0.02778} - 180 = 900 - 180 = 720\ \Omega$$

Final Answer

$R = 720\ \Omega$

Question 50

A galvanometer coil has a resistance of 15 Ω and it shows full-scale deflection for a current of 2.0 mA. How will you convert it into (a) an ammeter of range 0 to 5.0 A, (b) a voltmeter of range 0 to 15.0 V? Also determine the net resistances of the ammeter and the voltmeter.

★ **Likely Exam Question**

Given: $R_g = 15\ \Omega, \quad I_g = 2.0\ \text{mA} = 0.002\ \text{A}$

Formula: (a) $S = \dfrac{I_g R_g}{I - I_g}$, $R_{ammeter} = \dfrac{R_g S}{R_g + S}$ (b) $R = \dfrac{V}{I_g} - R_g$, $R_{voltmeter} = R_g + R$

Solution:

(a)

$$S = \frac{0.002 \times 15}{5 - 0.002} = \frac{0.03}{4.998} = 6.0 \times 10^{-3}\ \Omega$$

$$R_{ammeter} = \frac{15 \times 6.0 \times 10^{-3}}{15 + 6.0 \times 10^{-3}} = 5.99 \times 10^{-3}\ \Omega$$

(b)

$$R = \frac{15}{0.002} - 15 = 7500 - 15 = 7485\ \Omega$$

$$R_{voltmeter} = 15 + 7485 = 7500\ \Omega$$

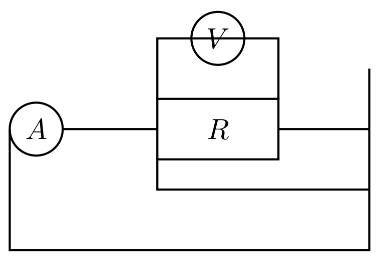
Final Answer

(a) Shunt $S = 6.0 \times 10^{-3}\ \Omega$, ammeter resistance $= 5.99 \times 10^{-3}\ \Omega$
(b) Series $R = 7485\ \Omega$, voltmeter resistance $= 7500\ \Omega$

Question 51

The given circuit is used to measure the resistance R. The ammeter reads 2 A and the voltmeter reads 120 V. What is the value of R if the resistance of the voltmeter is 3000 Ω? If the resistance of the voltmeter is taken to be infinitely large, then what is R?

★ **Likely Exam Question**



Circuit used to measure R: ammeter in the main line, voltmeter across R

Given: Ammeter reading $I = 2$ A, voltmeter reading $V = 120$ V, $R_v = 3000\ \Omega$

Formula: The voltmeter (in parallel with R) draws a current $I_v = \dfrac{V}{R_v}$; the actual current through R is $I_R = I - I_v$; then $R = \dfrac{V}{I_R}$.

Solution:

$$I_v = \frac{120}{3000} = 0.04 \text{ A}$$
$$I_R = 2 - 0.04 = 1.96 \text{ A}$$
$$R = \frac{120}{1.96} = 61.2 \ \Omega$$

If $R_v \to \infty$ (ideal voltmeter draws no current), then $I_R = I = 2$ A:

$$R = \frac{120}{2} = 60 \ \Omega$$

Final Answer

$R = 61.2\ \Omega$ (accounting for voltmeter current); $R = 60\ \Omega$ (ideal voltmeter)

Common Mistake: This is a classic ammeter-voltmeter method trap: since the ammeter here reads the *total* current (through R and through the voltmeter combined), you must subtract the voltmeter's own current before computing $R = V/I_R$ – simply dividing V/I gives a value close to, but not exactly equal to, the true resistance.